

K. Aditya Bharadwaj Nemana

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Education

PhD McGill University , Mechanical Engineering GPA: 3.9/4.0 • Relevant Courses: Turbulence, Numerical Analysis I, and Matrix Computations. • Thesis: Matrix-free preconditioners for Jacobian-free Newton-Krylov methods applied to turbulent flows. • Awards: Recipient of the McGill Engineering Doctoral Award (MEDA).	Montreal, Canada Sept 2021 – present
MSc McGill University , Mechanical Engineering GPA: 4.0/4.0 • Relevant Courses: Computational Aerodynamics, Computational Gas Dynamics, Turbomachinery and Propulsion, Dynamics of Combustion, and Finite Element Methods in CFD. • Thesis: Perturbation Signal Tracking Techniques for Unsteady Shock Wave Flows. • Awards: Recipient of the M. P. Paidoussis Prize for the best MSc thesis in the Department of Mechanical Engineering at McGill University in 2022.	Montreal, Canada Sept 2019 – Feb 2022
B.Tech National Institute of Technology Kurukshetra , Mechanical Engineering GPA: 9.45/10.0	Kurukshetra, India July 2014 – May 2018

Academic Experience

McGill University , Doctoral Candidate • Introduced a matrix-free preconditioner based on LU-SGS method in an in-house unstructured CFD code called FVENS using Eigen and PETSc libraries. • Worked closely with Bombardier Aerospace to improve the robustness of the Newton-Krylov solver in their unstructured CFD code, DRAGON. • Developed a matrix-free preconditioner based on the Newton-Schulz iterative method.	Montreal, Canada Sept 2021 – present
McGill University , Teaching Assistant • Conducted tutorials, graded assignments, and provided feedback to students. • Assisted with the development of course materials and resources.	Montreal, Canada Sept 2021 – present
Indian Institute of Technology (IIT) Kanpur , Research Associate • Worked on the design and development of a miniature power-generation system using liquid fuel. • Conducted computational fluid dynamics (CFD) analyses using ANSYS Fluent and ICEM CFD to evaluate system performance, efficiency, and optimize the heat-sink design.	Kanpur, India July 2018 – May 2019

Professional Experience

Bombardier Aerospace , Student Intern • Enhanced the robustness of DRAGON's Newton-Krylov solver by implementing a back-tracking line search and diagnosing and fixing critical bugs. • Implemented and tested a Jacobian-free Newton-Krylov framework with matrix-free preconditioning using PETSc library for applications in industrial RANS simulations.	Montreal, Canada Mar 2023 – present
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Post-Graduate Students' Society (PGSS), McGill University , Equity and Diversity Commissioner Accessibility Commissioner • Actively serving on McGill University's EDI Committee (EDIC) and co-chairing the PGSS Equity Committee. • Introduced EDI Awards aimed at supporting projects and initiatives that promote equity, diversity, and inclusion within the graduate student community. • Developed and implemented an accessibility survey for graduate students and postdoctoral researchers at McGill University.	Montreal, Canada May 2024 – present
Office of E-IDEA, McGill University , SCOPE Fellow • Member of the Social Impact, Community Building, and Outreach Partnerships in Engineering (SCOPE) team. • Designed and developed modules to help graduate students integrate EDI principles into engineering design.	Montreal, Canada Jan 2025 – present
Office of E-IDEA, McGill University , Summer Intern • Researched and compiled approaches to integrating EDI principles into engineering design. • Created and maintained a database of underrepresented engineers from marginalized communities and engineering innovations by indigenous peoples of Canada.	Montreal, Canada May 2024 – Aug 2024

Volunteering Experience

Subcommittee for Persons with Disabilities, McGill University , Co-Chair • Organized regular meetings to discuss and address the needs of persons with disabilities. • Represented the committee in meetings with university administration to advocate for the rights and needs of persons with disabilities.	Montreal, Canada Sept 2025 – present
Graduate Engineering Equity Committee (GEEC), McGill University , Events Coordinator • Organized monthly EDI-focused events, including movie nights, guest speaker talks, and panel discussions. • Managed event logistics, including registration forms, posters, room bookings, advertising, and feedback collection.	Montreal, Canada Mar 2022 – present
Council of Graduate and Postdoctoral Studies (CGPS), McGill University , Engineering Graduate Student Representative • Contributed to the development, revision, and approval of the policies on graduate and postdoctoral studies. • Participated in strategic planning for initiatives to promote and enhance graduate and postdoctoral studies.	Montreal, Canada Sept 2023 – Apr 2025

Skills

Programming and Software: Python, C/C++, Git, MATLAB, Bash, ANSYS Fluent, ICEM CFD, CATIA, FreeCAD, and MS Office.

Research Skills: Computational Fluid Dynamics (CFD), numerical methods, and applied mathematics.

Libraries and Frameworks: PETSc, Eigen, NumPy, SciPy, Matplotlib.

Languages: English, French (Beginner), Hindi, Telugu, and Odia.

Publications

On the disturbance tracking methods for shock wave reflection studies Nemana, Kameswara Aditya Bharadwaj, Timofeev, Evgeny 10.1063/5.0286049 (AIP Conference Proceedings)	Sept 2025
Development of a Standalone, Liquid-Fueled Miniature Power-Generation System Bharadwaj, N. K. A., Jain, N., Arghode, V. K. 10.1115/1.4044764 (Journal of Energy Resources and Technology)	Sept 2019